SYNTHETIC DATA — 교육용 가상 데이터이며 실제 제품 사양·운영 기준과 무관합니다.

Oil-immersed Transformer 1000kVA (TRF-OIL-1000) — Installation and Commissioning Guide

1. Safety

Installation, wiring and commissioning must be performed by qualified personnel. Disconnect and lock out all power sources before work. Verify absence of voltage.

2. Receiving and storage

Inspect the packaging on arrival and report damage within 7 days (see POL-RETURN). Store indoors in a dry place between -25 and +55 °C, relative humidity below 85% without condensation.

3. Mounting and clearances

- Required clearances: Minimum 500 mm from enclosure walls; maintain electrical clearances per IEC 61936-1
- Operating temperature: -25 to +40 °C
- Degree of protection: IP54 (outdoor)
- Above 1,000 m altitude, apply derating per the datasheet.

4. Wiring and tightening torques

- Terminal tightening torque: see dimension drawing
- Use copper conductors rated for at least 75 °C.
- Separate power and signal cables by at least 200 mm.

5. Commissioning procedure

1. Measure insulation resistance and polarisation index
2. Check off-circuit tap position against required secondary voltage
3. Verify temperature monitoring alarm and trip contacts
4. Check earthing of core and enclosure
5. Energise at no load and record secondary voltages

6. Pre-energisation checklist

☐ Product rating matches the drawing
☐ All terminals tightened to specified torque
☐ Protective earth connected
☐ Insulation resistance measured and recorded
☐ Settings / parameters recorded

7. Common commissioning issues

Symptom	Likely cause	See
Abnormal humming noise	Loose core clamping or harmonic load	TG-TRF-OIL-1000 §3.1
Buchholz alarm	Air trapped after oil filling	TG-TRF-OIL-1000 §3.2
Oil leakage at gasket	Gasket damage during transport	TG-TRF-OIL-1000 §3.4
Secondary voltage too high	Off-circuit tap at wrong position	TG-TRF-OIL-1000 §3.5

On-site commissioning support can be requested through the Global CS Center. For power transformers the request is handled by Field Service.